

Math 1301 Exam 1 Practice Questions

Name: _____

Section/Instructor: _____

Part 1. Consider the function $f(x) = x^x$.

Q1 $f'(x) =$

Q2 $f(x)$ is one-to-one on the intervals

Q3 $\int_1^e \ln(f(x))dx =$

Q4 (Challenge Problem) If $\int_1^e x^{x-1}dx = A$, then $\int_1^e x^x(1+\ln(x)) \ln(x)dx =$

(Your answer should be in terms of A)

Part 2. Calculate the following limits.

Q1 $\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln(x)} - \frac{1}{x-1} \right) =$

Q2 $\lim_{x \rightarrow 0^+} (\sin(x))^{\tan(x)} =$

Part 3. Calculate the following integrals.

Q1 $\int \frac{4^x + 2^x}{1 + 4^x} dx =$

Q2 $\int \sqrt{1 - x^2} \, dx =$

Q3 $\int_0^{\pi/5} x \cos(5x) \, dx =$

Q4 $\int \frac{e^{1/x} \cot(1 + e^{1/x})}{x^2} dx =$

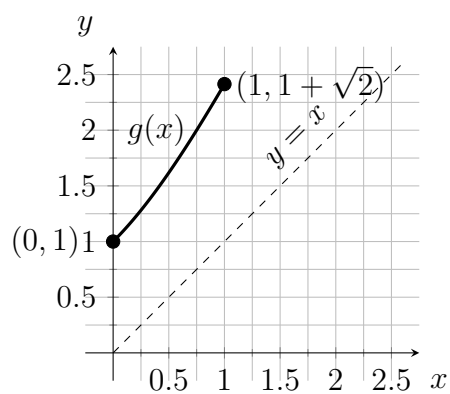
Q5 $\int \ln(1 + x^2) dx =$

Part 4. Consider the function $g(x) = x + \sqrt{1 + x^2}$.

Q1 $g(x)$ is invertible on

and $g^{-1}(x) =$

Q2 (Challenge Problem) $\int_0^1 g(x)dx =$



Part 5. Define the function $f(x) = \arctan(\ln(x+1)) - \arccos(\ln(\sqrt[4]{x^2+2x+1}))$

Q1 The domain of $f(x)$ is

Q2 $f'(x) =$

Q3 $f(x)$ has an inverse on

Q4 $f(e^{\sqrt{3}} - 1) =$